Quadratics in factored form

1

a

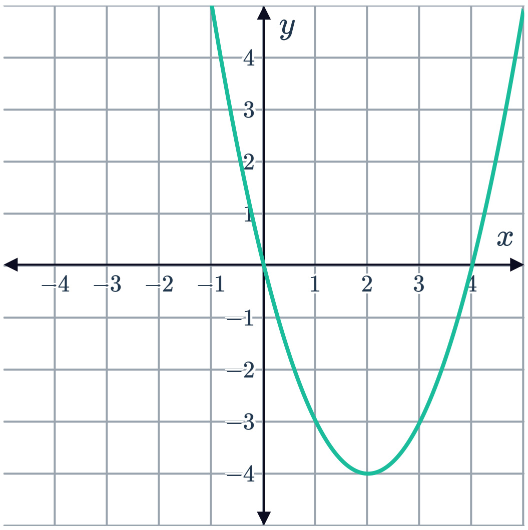
i $y=0$

ii $x=0, x=4$

iii $x=2$

iv $(2,-4)$

v



b

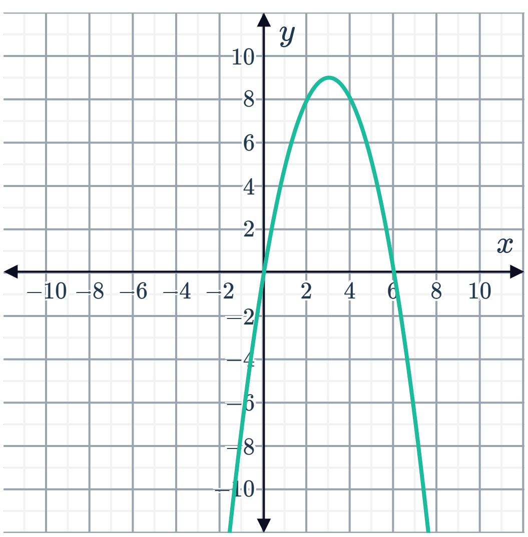
i $y=0$

ii $x=0, x=6$

iii $x=3$

iv $(3,9)$

v



c

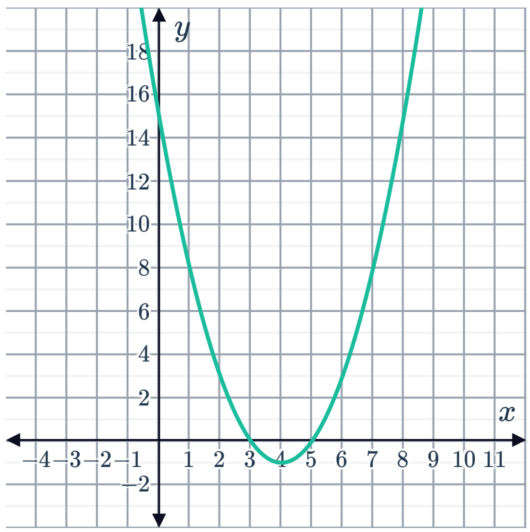
i $y = 15$

ii $x = 5, x = 3$

iii $x = 4$

iv $(4, -1)$

v



d

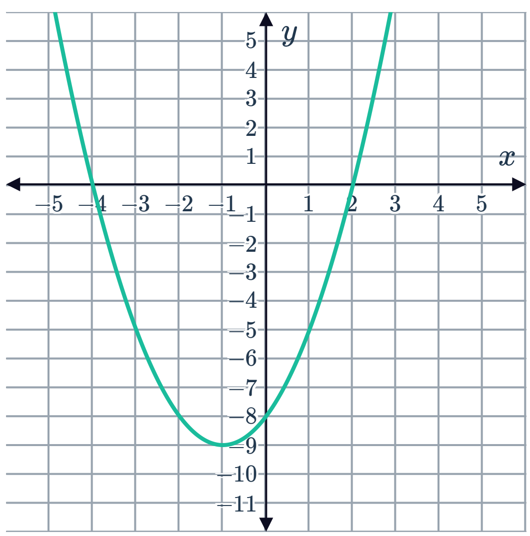
i $y = -8$

ii $x = -4, x = 2$

iii $x = -1$

iv $(-1, -9)$

v



e

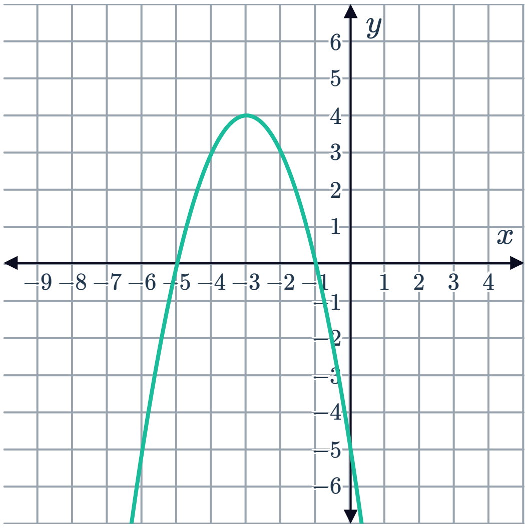
i $y=-5$

ii $x=1, x=5$

iii $x=3$

iv $(3,4)$

v



f

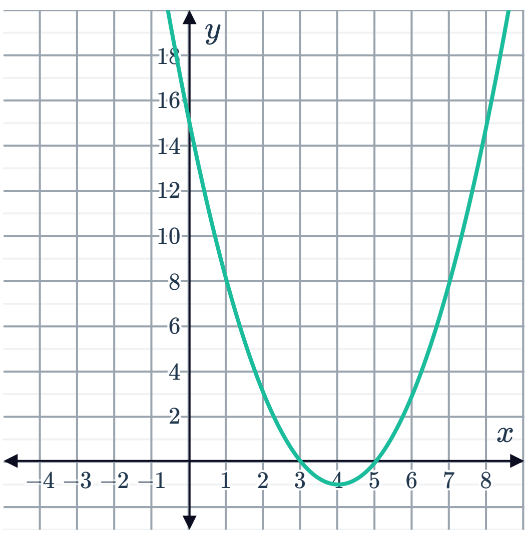
i $y=15$

ii $x=5, x=3$

iii $x=4$

iv $(4,-1)$

v



2

a

i 0

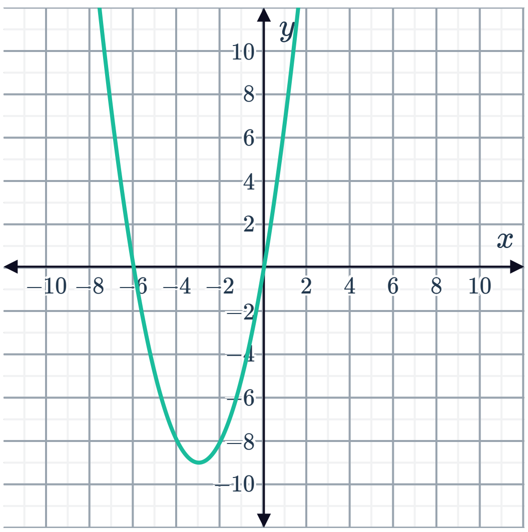
ii $x=0, x=-6$

iii

x	-5	-4	-3	-2	-1
y	-5	-8	-9	-8	-5

iv $(-3,-9)$

v



b

i 12

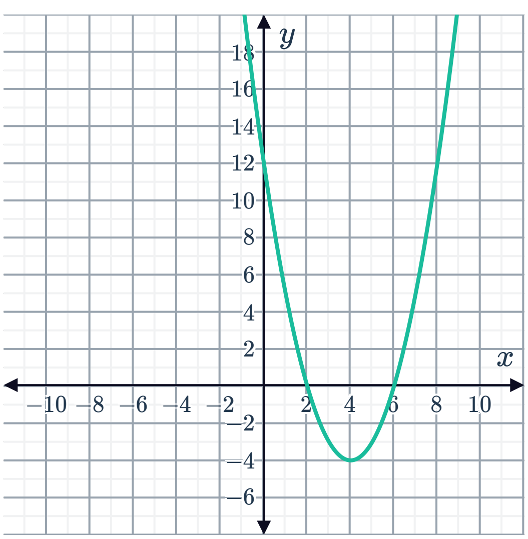
ii $x=2, x=6$

iii

x	0	2	4	6	8
y	12	0	-4	0	12

iv $(4,-4)$

v



c

i 8

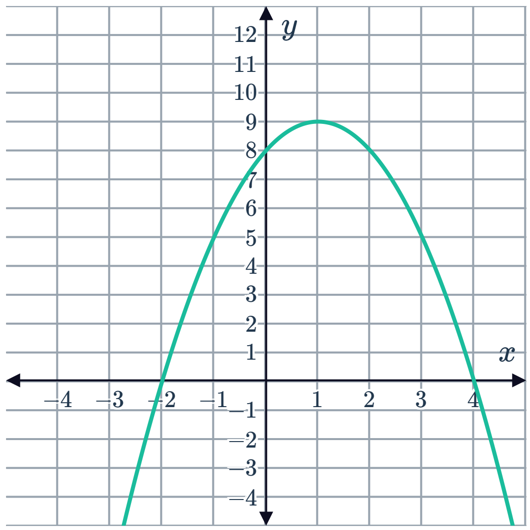
ii $x = 2, x = -4$

iii

x	-5	-3	-1	1	3
y	-7	5	9	5	-7

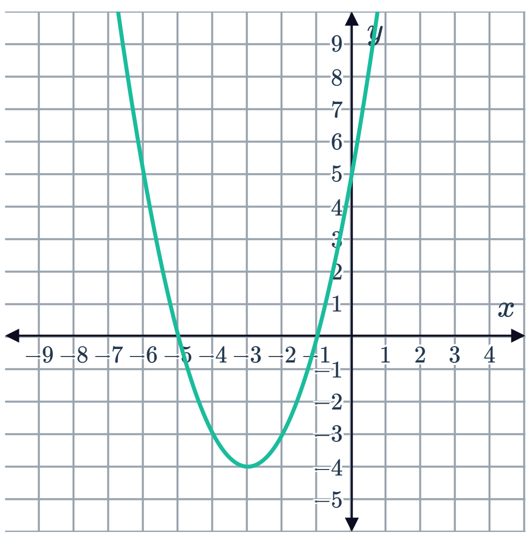
iv $(-1, 9)$

v



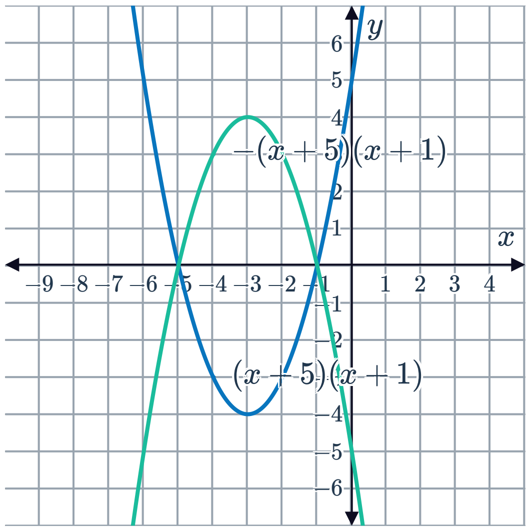
3

a



b -5

c



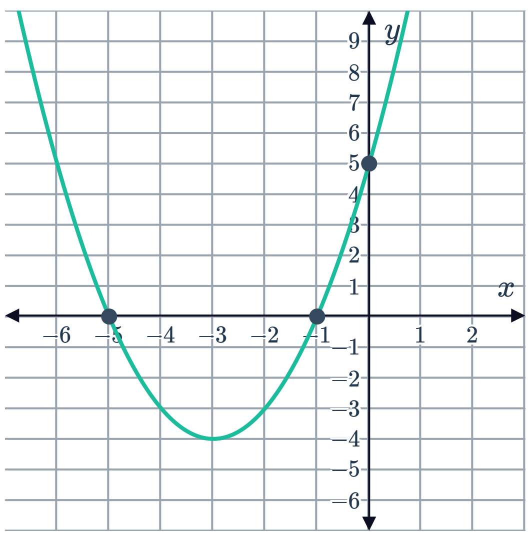
4

a **B**   b **C**   c **D**   d **A**

5

a $y=(x+1)(x+5)$ b $y=5$

c

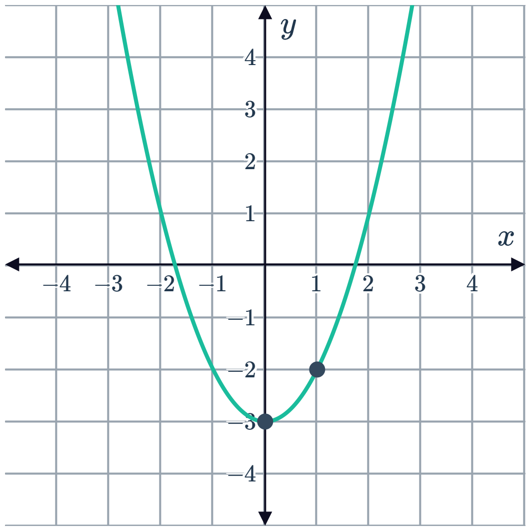


6

a $y = x^2 - 3$

b -2

c



7

a

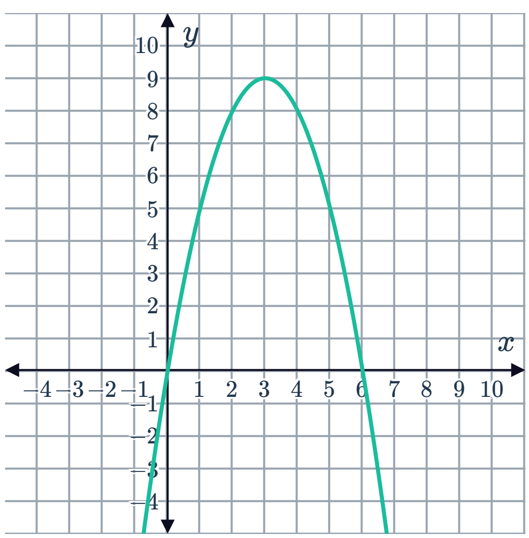
i $x\,(6 - x)$

ii $x = 0, x = 6$

iii 3

iv 9

v



b

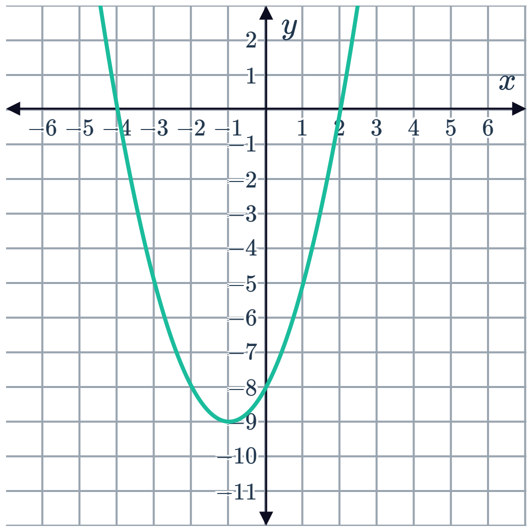
i $(x-2)(x+4)$

ii $x=2, x=-4$

iii -1

iv -9

v



8

a $x(4+x)$

b $x=0, x=-4$

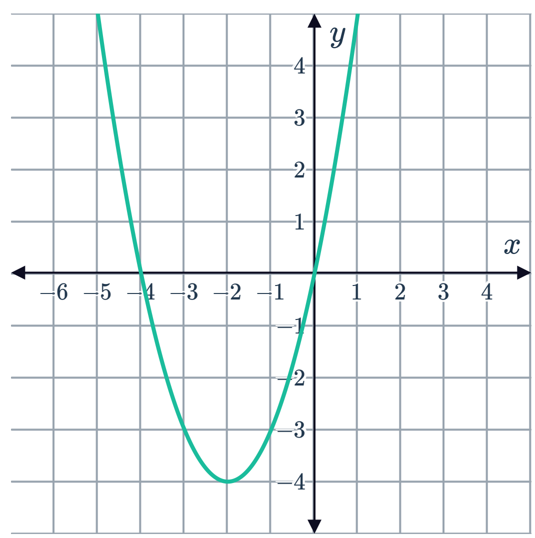
c 0

d

x	-4	-3	-2	-1	0
y	0	-3	-4	-3	0

e $(-2,-4)$

f



9

a $(x+2)(x+6)$

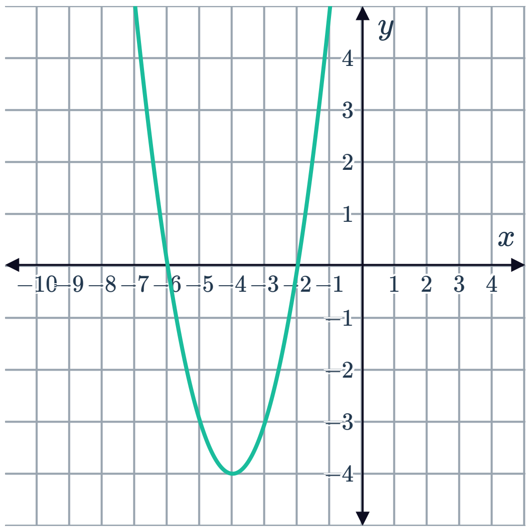
b $x=-2, x=-6$

c

x	-8	-6	-4	-2	0
y	12	0	-4	0	12

d $(-4,-4)$

e



10

a No

b Yes

11

a 8

b $y=-(x-2)(x+4)$

12 $x=0$

13

a $x=-11$

b $y=1(x-1)(x+11)$

c $(-5,-36)$

14

a $y=x^2-b^2$

b $a=0$

15

a $t=20$

b 100 m

Additional questions

16 $y = a(x - x_1)(x - x_2)$

The values of x_1 and x_2 are the x-values of the x-intercepts of the quadratic function. The scale factor of the function is represented by a.

17 The axis of symmetry of a quadratic is the vertical line that can be drawn so each side of the quadratic function mirrors the other. This line passes through the vertex of the quadratic function.

18

a $x = 1$ and 4

b $x = 3$ and -7

c $x = 0$ and 4

d $x = 1$ and -5

19

a $y = 24$

b $y = 0$

20 $x = 1$

21

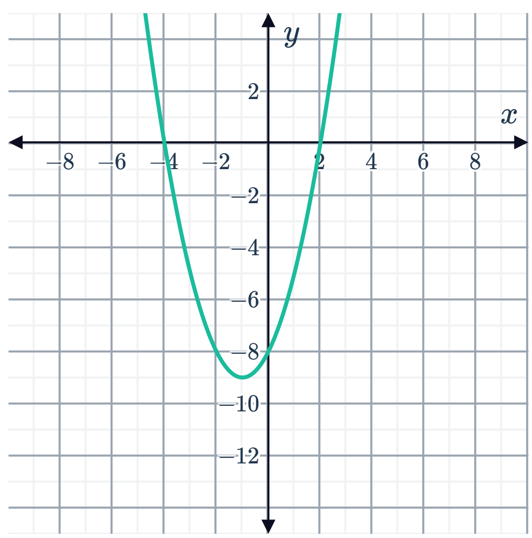
a

i $(2, 0)$ and $(-4, 0)$

ii $x = -1$

iii $(-1, -9)$

iv



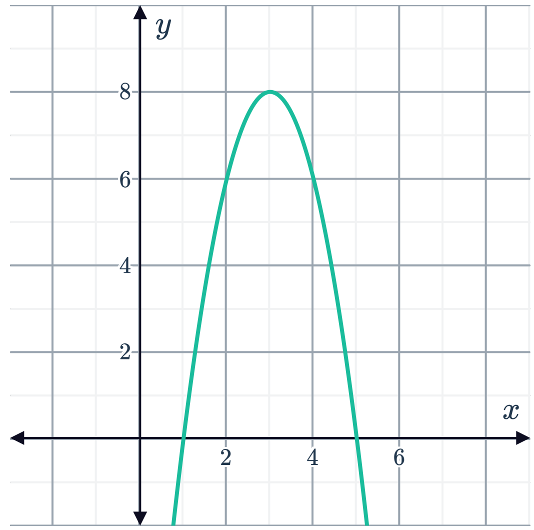
b

i $(1,0)$ and $(5,0)$

ii $x = 3$

iii $(3,8)$

iv



22

a

i $(1,0)$ and $(4,0)$

ii $(0,8)$

iii $y = 2(x-1)(x-4)$

b

i $(-2,0)$ and $(6,0)$

ii $(0,4)$

iii $y = -\frac{1}{3}(x-6)(x+2)$

23 $y = -4(x-5)(x+3)$

24 $y = \frac{1}{3}(x+2)(x-9)$

25 $y = 3(x+4)(x+7)$

26

a $(50, 110)$

b $y = -\frac{11}{250}x(x-100)$

c Domain: $0 \le x \le 100$
Range: $0 \le y \le 110$

27

a $(20,0)$ and $(40,0)$

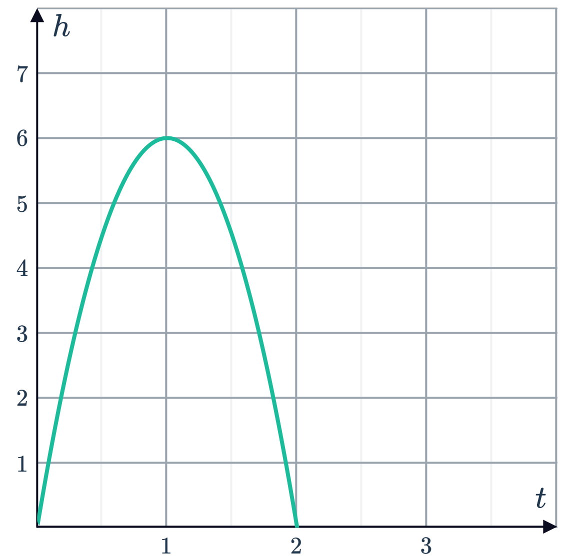
b $y = -\frac{1}{\;}(x-20)(x-40)$

28

a The x-intercepts represent the times when the eraser is 0 ft above Wilson's hand. This will be exactly when Wilson tosses the eraser and when it returns to his hand. Since Wilson tosses the eraser at $x = 0$ and it takes 2 seconds for it to return to his hand, the x-intercepts are $(0, 0)$ and $(2, 0)$. Since one of the x-intercepts is at the origin, we have the y-intercept is also $(0, 0)$.

b $(1, 6)$

c



d $y = -6x(x - 2)$

29 Since the vertex lies on the axis of symmetry, we know that 8 is the average of 12 and x_0. So then, we must have $x_0 = 4$.

If we substitute the coordinates of the vertex into the function, we get:

$$15 = k(8 - 4)(8 - 12)$$

If we solve this equation we get $k = -\dfrac{15}{16}$.